Issue	**1**
Factory ID	**25333**
Date of Issue	**13-Apr-2026**

Ali Trading Company (Pvt.) Ltd.

Pakistan ACCORD for Health and Safety in the Garment and Textile Industry

6 Km Daska Road
51310 Sialkot
Punjab Pakistan

Latitude: [32.45229], longitude: [74.49852]
Date of inspection: 12-Mar-2026

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 12-03-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

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1. Executive Summary

On 12-Mar-2026, Sarah Aiman and Azka Riaz on behalf of the Pakistan Accord carried out an initial electrical inspection for Ali Trading Company (Pvt.) Ltd., a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representative Mr. Syed Ghayyoor Abbas (GM Sustainability), Ms. Muqadas Shahzadi (EMS Manager), Mr. Aaqib Riaz(Electrical Engineer), Mr. Muhammad Rizwan (Structural Engineer), Mr. Muhammad Zeeshan (HR & Compliance Manager), and other staff members of the factory. The assessment was conducted as part of an overall Structural, Fire and Electrical initial inspection.

The factory complex is entirely occupied by Ali Trading Company (Pvt.) Ltd. as owner.
The factory operates in general shift with approximately 1500 workers out of which approximately 15% employees are female.

The factory comprises of 8 buildings and 12 ancillary structures. During the inspection, the areas highlighted in Figure 1.1 were surveyed, with the site boundary indicated in red dashed line.

- **Management Block Building** is a 2 storey B+G+1. The building is used for document store (basement), offices (ground floor), offices, kitchen, daycare, dining area (first floor), and an unoccupied roof.
- **Football Production 1 Building** is a 3 storey G+2. The building is used for press cutting, manual cutting, football issue, finish good warehouse (ground floor), football stitching, offices, thermos-bonded ball manufacturing, textile checking and textile packing (first floor), football printing & panel sorting (second floor), solar panels (roof).
- **Football Production 2 Building** is a 3 storey G+2. The building is used for raw material warehouse and bladder inspection (ground floor), football machine stitching and quality control (first floor), panel punching, bladder winding, stitcher printing and thermos bonding ball production (second floor), and an unoccupied roof.
- **Football Production 3 Building** is a 2 storey G+1. The building is used for washing, packing & final inspection (ground floor), molding, gluing, final stitch & bladder pasting (first floor), and solar panels (roof).
- **Composite & Textile Production Building** is a 2 storey. The building is used for paddle racket production, field hockey production, offices, chemical store, bag production, molding, pre-peg & paint booths (ground floor), storage (mezzanine 1), printing & offices (mezzanine 2), stitching, offices, cutting, packing & storage (first floor), and solar panels (roof).
- **Chemical Warehouse** is a two storey G+1. The building is used for storage (ground and first floor) and an unoccupied roof.
- **Composite Production 1** is a Ground only building. The building is used for composite semi finish, spray paint area, wooden hockey production, sanding, and storage (ground floor).
- **Composite Production 2** is a Ground only building. The building is used for padel racket drilling, composite semi finishing, and waste storage (ground floor).

Details of ancillaries are provided below:

- **Security room, purchase office and clinic** is a Ground only structure. The structure is used for Security room, purchase office and clinic.
- **Electrical workshop and driver room** is a Ground only. The structure is used for Security room, purchase office and clinic.
- **Mosque** is a Ground only structure. The structure is used for prayer area.
- **Generator room 1** is a Ground only structure. The structure is used for generators and electric panels.
- **Generator room 2 and compressor room** is a Ground only structure. The structure is used for generators, electric panels and compressors.
- **Security guard room** is a Ground only structure. The structure is used for security guard residencies.
- **Canteen** is Ground only structure. The structure is used for dining area.
- **Cooking area** is Ground only structure. The structure is used for dining area.
- **R & D Lab** is a Ground only structure. The structure is used for laboratory.
- **Press cutting hall** is a Ground only structure. The structure is used for cutting operation.
- **Lamination hall** is a Ground only structure. The structure is used for lamination operation.
- **Welding workshop** is a Ground only steel shed. The structure is used for welding operation.

A summary of the main concerns in relation to life safety systems identified during the inspection (non-exhaustive) that should be actioned is as follows:

- **Abnormal Temperature:** Within electrical panels, abnormal temperatures were recorded at various locations (refer to section 7.2 for further details). Hotspots within panels can cause short circuits and fire due to insulation damage and overheating of cables or electrical components.
- **Fire Alarm and Detection System:** Two conventional fire alarm control panels were installed to serve the entire facility. Several deficiencies were identified in the fire alarm and detection system, including fault indication on the security office panel, faulty manual call points and sounders, missing or improperly placed manual call points, inadequate detection coverage, zero to low audibility in certain areas, and the use of non-fire-rated cables, posing a significant life safety risk in the event of a fire.
- **Emergency lighting System:** The emergency lighting system had inadequate coverage along evacuation routes, with non-functional fixtures, insufficient battery backup, and non-illuminated directional signage in various areas of the facility, hindering safe evacuation during power outages and increasing the risk of accidents.
- **Earthing system:** The earthing system was found to be inadequate in several areas of the facility. Some distribution boards lacked earthing conductors, and many electrical machines were not provided with proper earth connections. The lack of a proper earthing system poses significant safety risks, such as electric shocks, fire hazards, and potential equipment damage.

We see no reason to suspend operations in the facility due to these concerns, subject to the actions identified being addressed within the specified timeframes.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents



Figure 1.1. Site plan (Satellite view) – Red dashed line indicates site boundary, highlighted areas indicates buildings inspected

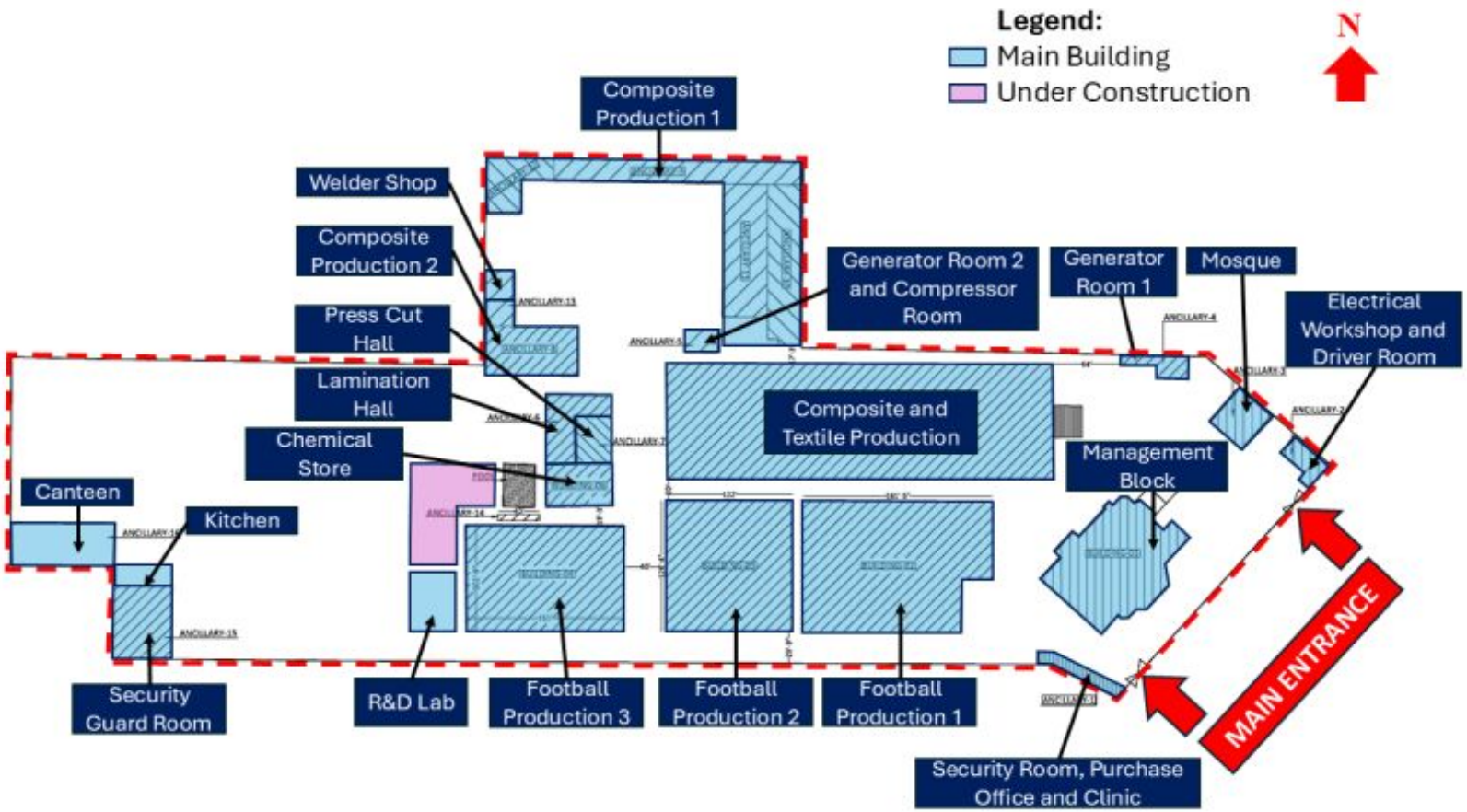


Figure 1.2. Site layout plan of Ali Trading Company (Pvt.) Ltd.

2.1 Building Use

The factory comprises eight main buildings and twelve ancillary structures. The floor-wise activity details are as follows:

1) Management Block Building
B00 – Basement: Document store.
L00 – Ground Floor: Offices.
L01 – First Floor: Offices, kitchen, daycare, and dining area.
L99 – Roof: Unoccupied.

2) Football Production 1 Building
L00 – Ground Floor: Press cutting, manual cutting, football issue, and finished goods warehouse.
L01 – First Floor: Football stitching, offices, thermos-bonded ball manufacturing, textile checking, and packing.
L02 – Second Floor: Football printing and panel sorting.
L99 – Roof: Solar panels.

3) Football Production 2 Building
L00 – Ground Floor: Raw material warehouse and bladder inspection.
L01 – First Floor: Football machine stitching and quality control.
L02 – Second Floor: Panel punching, bladder winding, stitcher printing, and thermos bonding ball production.
L99 – Roof: Unoccupied.

4) Football Production 3 Building
L00 – Ground Floor: Washing, packing, and final inspection.
L01 – First Floor: Molding, gluing, final stitching, and bladder pasting.
L99 – Roof: Solar panels.

5) Composite & Textile Production Building
L00 – Ground Floor: Paddle racket production, field hockey production, offices, chemical store, bag production, molding, prepreg, and paint booths.
M01 – Mezzanine 1: Storage.
M02 – Mezzanine 2: Printing, offices.
L01 – First Floor: Stitching, offices, cutting, packing, and storage.
L99 – Roof: Solar panels.

6) Chemical Warehouse
L00 – Ground Floor: Storage.
L01 – First Floor: Storage.
L99 – Roof: Unoccupied.

7) Composite Production 1
L00 – Ground Floor: Composite semi-finish, spray paint area, wooden hockey production, sanding, and storage.

8) Composite Production 2

L00 – Ground Floor: Padel racket drilling, composite semi-finishing, and waste storage.

Ancillary Structure:

Twelve ancillary structures are as follows:

1) Security Room, Purchase Office & Clinic

L00 – Ground Floor: Security room, purchase office, and clinic.

2) Electrical Workshop & Driver Room

L00 – Ground Floor: Electrical workshop, and driver room.

3) Mosque

L00 – Ground Floor: Prayer area.

4) Generator Room 1

L00 – Ground Floor: Generators and electrical panels.

5) Generator Room 2 & Compressor Room

L00 – Ground Floor: Generators, electrical panels, and compressors.

6) Security Guard Room

L00 – Ground Floor: Security guard residence.

7) Canteen

L00 – Ground Floor: Dining area.

8) Cooking Area

L00 – Ground Floor: Cooking area.

9) R&D Lab

L00 – Ground Floor: Laboratory.

10) Press Cutting Hall

L00 – Ground Floor: Cutting operations.

11) Lamination Hall

L00 – Ground Floor: Lamination operations.

12) Welding Workshop

L00 – Ground Floor: Welding operations.

2.2 Utility Source

The facility receives a single 11 kV utility supply from Gujranwala Electric Power Company (GEPCO) as its primary power source, which is received at the Electrical Workshop. The Electrical Workshop houses the HT panel, a 630 kVA transformer, and LT panels supplying the football production buildings. A second 630 kVA transformer with LT panels is located in Generator Room 2, serving the textile and composite production areas. Low-voltage supply from the LT panels is distributed throughout the facility via underground cabling to multiple distribution boards. Additionally, a 400 kW solar power system is synchronized with the primary utility supply.

2.3 Backup source

The facility has four low-voltage diesel generators—two rated at 200 kVA, one at 250 kVA, and one at 550 kVA—located in Generator Rooms 1 and 2, serving as backup power sources. The fire alarm and detection system, along with the emergency lighting system, are equipped with built-in battery backup supplies.

3. Documentation

This table shows the details of the documentation provided by the factory.

S.NO	Description	Available (Y/N)	Created Date	Status / Remarks
1	Electrical Layout Drawing	No		
2	Single Line Diagrams	No		
3	Fire Alarm Drawing	No		
4	Earth Test Report	No		
5	Thermographic Test Report	No		
6	Insulation Resistance Test Report	No		
7	LPS Risk Assessments	No		
8	Periodic Maintenance Record	No		
9	Transformer Oil Testing Report	No		

4. Electrical Safety Measures

During our electrical site inspection of Ali Trading Company (Pvt.) Ltd., a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within Ali Trading Company (Pvt.) Ltd.. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a licensed professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202603-064126

Timeline	Remediation Plan
Medium Term CAP-0075243	Factories should begin remediation of CAP items such as cleaning, cable organisation, and basic corrections using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Engineer is required.

6. Diagrams & Drawings

6.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no substation room layout drawings available for the factory.

There are no visible indications of substation equipment locations or pathways for maintenance and emergency access.

The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202603-064097

Timeline	Remediation Plan
Medium Term CAP-0075443	Develop detailed substation room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0075444	Regularly update and review the substation room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

6.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were missing.

This deficiency was noted in all electrical rooms and areas where SLDs should be prominently displayed.

The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202603-064090

Timeline	Remediation Plan
Medium Term CAP-0075445	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0075446	Implement a regular review and update process for the single-line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked electrical layout drawings - including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202603-064091

Timeline	Remediation Plan
Medium Term CAP-0075448	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0075449	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.

During the survey, the circuit breakers installed throughout the facility, inside the electrical panels and distribution boards, were inspected.

i. It was observed at some locations that circuit breakers were damaged and burned out. Damaged and burned-out circuit breakers pose a serious fire hazard and risk of electrical shock due to failure to interrupt fault currents and accidental touch. This issue was specifically observed in the textile stitching distribution board on the first floor of the Composite & Textile Production Building.

ii. It was found in the distribution board on the second floor of Football Production 2 Building that a single-phase load was fed from a three-pole breaker. Using a single phase of a three-pole breaker can lead to unbalanced load conditions. This can cause the breaker to trip unexpectedly or, in some cases, fail to trip when necessary, compromising the safety of the system.

iii. It was observed that the circuit breakers were installed in the Electrical Workshop and General Waste Area without proper fixing supports and protective enclosures. This inadequate fixing could lead to poor connections, arcing, short circuits, or other electrical faults.

iv. It was also found that the circuit breaker was installed in outdoor areas without weather-protective enclosures, posing a risk of short circuits or electrical incidents during adverse weather conditions. This issue was specifically observed outside Generator Rooms 1 & 2, outside the Football Production 1 Building, outside the General Waste Area, and in the Gas Steamers Area.

v. It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes.
a. In the LT panel in Generator Room 1, a 320 A circuit breaker was connected to 50 mm^2 PVC cables, and a 63 A circuit breaker was connected to 4 mm^2 PVC cables.
b. In the LT panel in Generator Room 2, a 100 A circuit breaker was connected to 2.5 mm^2 PVC cables.
c. In the cutting distribution board on the ground floor of Football Production 1 Building, a 100 A circuit breaker was connected to 6 mm^2 PVC cables.
d. In the distribution board on the second floor of Football Production 2 Building, a 60 A circuit breaker was connected to 4 mm^2 PVC cables.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and

potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202603-064084

Timeline	Remediation Plan
Medium Term CAP-0075506	i. Replace all damaged and burned-out circuit breakers with properly rated units and ensure proper installation and regular inspection to maintain safe operation.
Medium Term CAP-0075524	ii. Remove single-phase connections from three-pole breakers and replace them with appropriate single-pole breakers, ensuring compliance with standard current-carrying capacities and preventing load imbalance.
Medium Term CAP-0075525	iii. Secure all loosely installed circuit breakers with proper mounting and enclose them in compliant metallic or insulated enclosures, in accordance with BS 7671 Section 416 (Barriers and Enclosures for Basic Protection).
Medium Term CAP-0075526	iv. Relocate circuit breaker boxes to indoor or sheltered areas. If relocation is not feasible, install weatherproof enclosures with a suitable IP rating for outdoor conditions, as per BS 7671 Clause 512.2.
Medium Term CAP-0075527	v. Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - 320 A circuit breaker connected to 50 mm² PVC cables in LT panel Generator Room 1



Figure 2.2 - Circuit breakers found damaged and burned out in the textile stitching distribution board of Composite & Textile Production



Figure 2.3 - Circuit breaker was found unsupported inside hockey distribution board Composite Production 1



Figure 2.4 - Circuit breaker was found unsupported and without protective enclosure in Electrical Workshop



Figure 2.5 - Circuit breaker installed outside Generator Room 1 with non-IP rated enclosure



Figure 2.6 - 60 A circuit breaker connected to 4 mm² PVC cables and a single phase load fed from three pole breaker Football Production 2 distribution board

7.2 Hot Spots

This relates to the entire facility.

During the inspection, Thermographic surveys were conducted on the electrical boards.

The survey identified numerous hot spots within panels and distribution boards. Hot spots in electrical distribution systems can be caused by several factors, including:

> Loose connections
> Overloaded circuits
> Faulty components
> Improper sized cable
> Insulation degradation

It was also found that the thermographic report was not available at the site. Without thermography testing, faults such as overheating components or loose connections may go unnoticed, which increases the risk of fire. Early detection through thermography can prevent fire incidents.

The categories of identified hot spots have been evaluated as per the ANSI NETA MTS 2019 standard, Table 100.18. Hot spots were observed, particularly at the following locations:

i. 235°C at the cable inside the cable tray for solar inverters located in the Generator Room 1.
ii. 150°C at neutral bar cable connection on the first-floor textile stitching distribution board in Football Production 1.
iii. 146°C at the circuit breaker cable connection on the first-floor molding machine distribution board in Football Production 3.
iv. 111°C at the red phase busbar connection in the LT panel located in the Generator Room 2.
v. 98°C, 85°C, and 83°C at the circuit breaker cable connection in the LT panel located in the Generator Room 2.
vi. 88°C at the circuit breaker cable connection and 73°C at the red phase cable in the distribution board on the second floor of the football printing in Football Production 1.
vii. 83°C at the circuit breaker cable connection on the second floor of the textile printing in Football Production 2.
viii. 83°C at the circuit breaker cable connection in the main distribution board on the ground floor in the Composite & Textile Production Building.
ix. 79°C at the red phase busbar connection and 78°C at the yellow phase busbar connection in the LT panel located in Generator Room 2.

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC-insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature event that can cause severe injuries or fatalities to personnel working near the affected equipment.F-202603-064080

Timeline	Remediation Plan
Immediate CAP-0075480	Any circuits identified on site with temperatures exceeding 40°C above the ambient temperature, operating above 90°C or exhibiting visual signs of burning must be immediately isolated and taken out of service. These circuits should remain de-energized until a thorough review has been conducted and appropriate remedial actions have been implemented to resolve the underlying issues.
Medium Term CAP-0075481	During the inspection, several circuits were identified where the temperature was exceeding 20°C above ambient temperature. It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term CAP-0075482	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating.

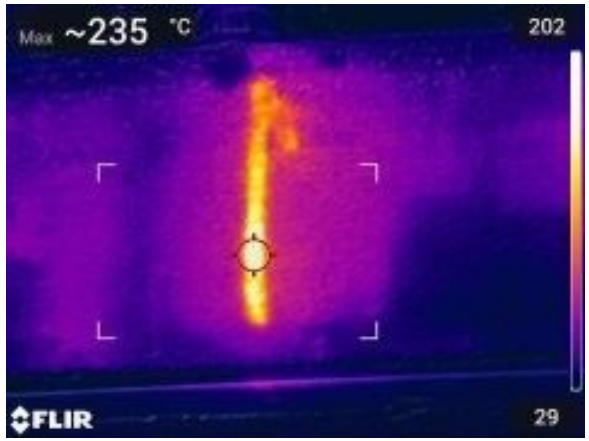


Figure 2.7 - Hot spot found on the cable inside the cable tray for solar inverters located in the Generator Room 1



Figure 2.8 - Hot spot found on the cable inside the cable tray for solar inverters located in the Generator Room 1



Figure 2.9 - Hot spot found on the neutral bar cable connection on the first-floor textile stitching distribution board in Football Production 1



Figure 2.10 - Hot spot found on the neutral bar cable connection on the first-floor textile stitching distribution board in Football Production 1

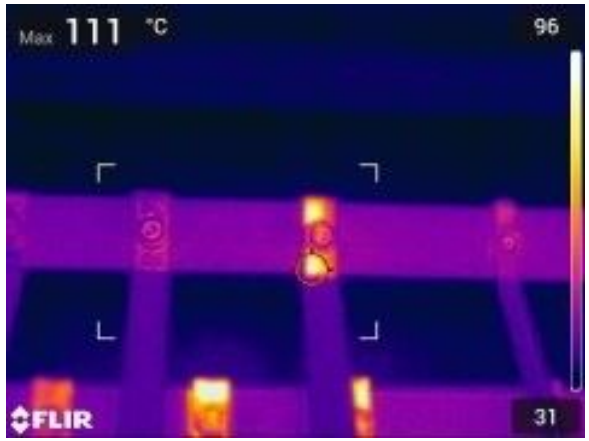


Figure 2.11 - Hot spot found on the red phase busbar connection in the LT panel located in the Generator Room 2

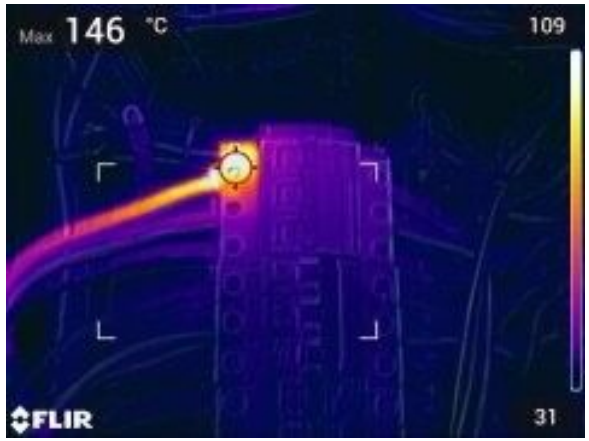


Figure 2.13 - Hot spot found on the circuit breaker cable connection on the first-floor molding machine distribution board in Football Production 3



Figure 2.12 - Hot spot found on the red phase busbar connection in the LT panel located in the Generator Room 2



Figure 2.14 - Hot spot found on the circuit breaker cable connection on the first-floor molding machine distribution board in Football Production 3

7.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

i. No earthing system design documentation or records of earth pit testing and maintenance were provided by the factory, indicating a lack of verification of the earthing system’s compliance and ongoing performance in accordance with relevant standards.

ii. The earth pits were inaccessible and unsuitably located. This issue was observed outside the Electrical Workshop and Generator Room 2.

iii. The earthing system was absent in multiple locations. Several distribution boards lacked earthing cables, and many electrical machines did not have proper earth connections. This issue was specifically observed inside the distribution boards of Composite Production 1 & 2 Buildings, textile stitching machines in Composite & Textile Production Building, textile printing distribution board in Football Production 2 Building, main distribution board on the first floor in Football Production 3 Building, sand blast machine distribution board, canteen distribution board, as well as electric water chillers in Football Production 1 Building.

iv. Equipotential bonding for the cable trays and metallic panel covers was missing throughout the facility.

The lack of a proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202603-064085

Timeline	Remediation Plan
Medium Term CAP-0075501	i. Develop a complete earthing system design for the facility and prepare earthing layout drawings in accordance with BS 7671 (18th Edition + A2:2022) Chapter 54. Submit the earthing system design and drawings to Pakistan Accord for review and approval.
Medium Term CAP-0075502	ii. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54.

Medium Term CAP-0075503	iii. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1.
Medium Term CAP-0075504	iv. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.
Longer Term CAP-0075505	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.15 - Earth connection missing for electric water chiller in Football Production 1



Figure 2.16 - Earth connection missing in the textile printing distribution board in Football Production 2



Figure 2.17 - Equipotential bonding missing for the cable trays in Football Production 1 Building



Figure 2.19 - Earth connection missing in the main distribution board in Composite Production 1



Figure 2.18 - Earthing pit inaccessible outside the Electrical Workshop



Figure 2.20 - Earth connection missing in the main distribution board in Football Production 3

7.4 Panel/cabling/busbar

This relates to the entire facility.

The visual site inspection included an examination of the electrical panels, cables, and busbars throughout the facility.

i. It was observed that the electrical panel in the Composite Production 2 Building was installed on a wooden support, posing a fire risk due to the combustible nature of the material and potential overheating.

ii. Several electrical panels exhibited signs of corrosion, indicating potential exposure to moisture or other corrosive elements. This issue was observed in the distribution boards located on the first floor of Football Production 1 Building and the electrical enclosure outside Generator Room 2.

iii. There were visible signs of damage to panels, cables, and busbars, including broken panel locks, burnt cable insulations, and non-operational voltmeters and ammeters, etc. This issue was observed in the distribution boards located in the Football Production 2, Composite Production 1, and Composite & Textile Production Buildings.

iv. Phase separators were missing in the LT panels and most of the distribution boards, posing the risk of accidental contact between phases.

v. Inadequate cable dressing was observed in most of the distribution boards, with conductors not properly arranged, secured, or terminated neatly. This resulted in untidy wiring, poor segregation between circuits, and unnecessary strain on terminations, which can lead to loose connections, overheating, and difficulty during inspection or fault-finding. This issue was specifically observed in the LT panels in the Generator Room 2 and the hockey distribution board in Composite Production 1.

The utilization of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202603-064083

Timeline	Remediation Plan
Immediate CAP-0075528	i. Install the electrical panel on a non-combustible support and remove the wooden backing.

Medium Term CAP-0075529	ii. Replace or refurbish corroded electrical panels to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials.
Medium Term CAP-0075530	iii. Inspect and repair or replace all damaged panels, burnt circuit breakers, damaged cable insulation, and faulty instruments to ensure the integrity of the electrical system, in accordance with Section 522 of BS 7671 18th Edition + A2 (2022).
Medium Term CAP-0075531	iv. Phase separators should be installed between adjacent live conductors of different phases within distribution boards to ensure safe segregation and protection against direct contact, in accordance with BS 7671 Regulation 416.2.
Medium Term CAP-0075532	v. Ensure proper cable management by organizing and securing cables within the panels using cable ties, clips, or ducting. Verify that all components and connections are securely mounted, neatly arranged, and labelled to improve safety, ease of maintenance, and compliance with electrical standards such as BS 7671.
Longer Term CAP-0075852	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.





Figure 2.21 - Burnt cable insulation observed in the hockey distribution board, Composite Production 1



Figure 2.22 - Electrical panel installed on a wooden support in Composite Production 2 Building



Figure 2.23 - Phase separators missing and inadequate cable dressing observed in the LT panel, Generator Room 2



Figure 2.24 - Panel lock broken of the football production & textile lighting distribution board



Figure 2.25 - Improper cable dressing with PVC conduits and exposed busbars observed inside the hockey distribution board, Composite Production 2

Figure 2.26 - Corrosion observed in textile stitching distribution board, Football Production 2



Figure 2.27 - Non-functional voltmeter and ammeter observed in football stitching distribution board, Football Production 2



Figure 2.28 - Corrosion observed on electrical enclosure outside Generator Room 2

7.5 Loop Connection

This relates to the entire facility.

During the inspection, loop connections were identified within the electrical panels. This issue was specifically observed in the distribution boards located in Composite Production 1 & 2 Buildings, Composite & Textile Production Building, Football Production 1 & 2 Buildings, and the Canteen.

In these loop connections, one circuit breaker is energized by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202603-064079

Timeline	Remediation Plan
Medium Term CAP-0075476	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilized to provide each circuit with the required live, neutral, and earth connections.



Figure 2.29 - Loop connections found in the heat transfer machines distribution board, Composite & Textile Production



Figure 2.31 - Loop connections found in the warehouse distribution board, Football Production 2



Figure 2.30 - Loop connections found in the cutting distribution distribution board, Football Production 1



Figure 2.32 - Loop connections found in the main distribution board, Composite Production 1

7.6 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility. This issue was specifically observed in the LT panel and lighting distribution board of the Generator Room 1, the compressor distribution board of the Generator Room 2, and the sand blast machine distribution board.

In these cases, multiple circuits were being protected by a single circuit breaker.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit.F-202603-064092

Timeline	Remediation Plan
Medium Term CAP-0075479	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022).



Figure 2.33 - Multiple connections found in the lighting distribution board, Generator Room 1



Figure 2.34 - Multiple connections found in the LT panel, Generator Room 1



Figure 2.35 - Multiple connections found in the compressor distribution board, Generator Room 2

7.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

Unsealed openings and missing baseplates were observed in the panels and distribution boards. This issue was specifically observed in the LT panel located in the Electrical Workshop, the LT panels and solar inverter panels in Generator Room 1, the LT panels in Generator Room 2, the football molding distribution board in Football Production 3, the Football Manual Cutting distribution board, and the Purchase Office electrical enclosure.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202603-064086

Timeline	Remediation Plan
Medium Term CAP-0075499	Corrective actions are required to seal all openings and replace missing panels to restore the integrity and safety of the electrical systems. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0075500	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.36 - Unsealed sealed openings observed in the solar inverter panel in Generator Room 1



Figure 2.37 - Unsealed sealed openings observed in the football molding distribution board in Football Production 3



Figure 2.38 - Unsealed openings observed in the LT panel in Electrical Workshop



Figure 2.39 - Unsealed openings observed in the LT panel in Generator Room 1

7.8 Adequate accessibility

This relates to the entire facility.

During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

At numerous locations, it was found that the electrical panels do not possess the required space for inspection or easy access to the panels and their components. This issue was specifically observed in the Management Building near the main entrance, on the first floor of Football Production 1, on the first floor of Composite & Textile Production Building, and Composite Production 1 & 2 buildings.

Limited accessibility to distribution panels and machine panels hampers effective maintenance and inspection procedures. This can potentially cause delays in addressing emergencies and troubleshooting electrical problems. These accessibility challenges could compromise the operational reliability and safety of the premises, as prompt maintenance and emergency responses are hindered. F-202603-064088

Timeline	Remediation Plan
Medium Term CAP-0075483	Electrical equipment shall be arranged to provide sufficient space for the initial installation and later replacement of individual items of electrical equipment. Sufficient allowance shall also be made for sufficient space for the initial installation and later replacement of individual items of electrical equipment in accordance with Section 132.12 of BS7671 18th Edition + A2 (2022).



Figure 2.40 - Inadequate accessibility for the distribution board on the first floor of Football Production 1



Figure 2.41 - Inadequate accessibility for the distribution board, Management Building



Figure 2.42 - Inadequate accessibility for the distribution board on the first floor of Football Production 1



Figure 2.43 - Inadequate accessibility for the stitching distribution board on the first floor of Composite & Textile Production



Figure 2.44 - Inadequate accessibility for the thermo mall distribution board on the second floor of Football Production 2



Figure 2.45 - Inadequate accessibility for the distribution board in the Composite Production 2

7.9 Lint & dust inside panel board

This relates to the entire facility.

During the inspection, the cleanliness of electrical panels, distribution boards, and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in most of the distribution boards, panels, and trunking systems. This issue was specifically observed in the LT panel located in the Electrical Workshop, compressor distribution board in Generator Room 2, warehouse distribution board in Football Production 2, CNC drilling panel in Composite Production 2, waste water pump electrical enclosure outside Football Production 3, and compressor distribution board in the gas steamer area.

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating.

Additionally, the accumulation of dust and lint can obstruct proper operation and maintenance, thereby impairing the functionality of the electrical components.F-202603-064087

Timeline	Remediation Plan
Medium Term CAP-0075497	All electrical panels and trunking runs shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022).
Longer Term CAP-0075498	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and trunking runs. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.46 - Dust observed in the LT panel located in the Electrical Workshop



Figure 2.47 - Dust and debris found in the waste water pump electrical enclosure outside Football Production 3



Figure 2.48 - Lint and dust found in the compressor distribution board in Generator Room 2



Figure 2.49 - Dust observed in the warehouse distribution board in Football Production 2

7.10 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

It was observed that a lightning protection system was provided; however, there was no study available to confirm its adequacy, and no risk assessment was available.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202603-064093

Timeline	Remediation Plan
Medium Term CAP-0075477	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment according to BS EN 62305 (Protection against Lightning) standard. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0075478	If the risk assessment concludes that an LPS is necessary, then hire a qualified engineer to perform an adequacy study of the existing LPS. This study should evaluate whether the current system meets the necessary design and performance standards according to BS EN 62305. If the study finds that the current LPS is inadequate, make the necessary upgrades to bring the system up to standard.

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

Two conventional fire alarm control panels were installed to serve the entire facility, one in the Security Room and the other near the Composite Production 2 building. Several deficiencies were identified in the fire alarm and detection system, including fault indication on the security office panel, faulty manual call points and sounders, missing or improperly placed manual call points, inadequate detection coverage, zero to low audibility in certain areas, and the use of non-fire-rated cables, posing a significant life safety risk in the event of a fire.

The fire alarm and detection system testing revealed the following observations:

- Zero audibility in sewing line 8 on the first floor of the Football Production 1 Building.
- Low audibility in the store on the ground floor, sewing lines on the first floor, and zero audibility in the punching hall on the second floor of the Football Production 2 Building.
- Low audibility and insufficient visual coverage on the first floor of the Football Production 3 Building.
- Low audibility on the mezzanine and faulty sounders on the first floor of the Composite & Textile Production Building.
- Faulty sounders in the canteen, lamination hall, and press cut hall ancillary structures.

Listed panels & components are tested and verified to meet specific safety and performance standards. Non-listed panels may not provide reliable detection and alarm functions, which can compromise timely evacuation and firefighting efforts. Conventional fire alarm systems may lack advanced features such as individual device identification and precise location of alarms, which can delay response times during emergencies.

A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm. A delayed response during a fire event can significantly increase the impact on personnel, goods, and property.F-202603-064099

Timeline	Remediation Plan

Immediate CAP-0075266	Ensure the functionality and performance of the existing alarm and detection system, and remove the fault indicated on the security office fire alarm control panel .
Medium Term CAP-0075267	i. Submit design documentation for the new replacement of FADS system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Longer Term CAP-0075268	ii. Submit documentation to the Pakistan Accord showing that an L/C has been opened for import of any components required for the FADS design submitted earlier.
Longer Term CAP-0075269	iii. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation.
Longer Term CAP-0075270	iv. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review.
Longer Term CAP-0075271	v. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.
Longer Term CAP-0075272	vi. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.50 - Detection coverage not provided above the ceiling within the void space of the Football Production 2 Building



Figure 2.52 - Insufficient visual coverage on the first floor of Football Production 3 Building



Figure 2.51 - Conventional fire alarm control panel installed near Composite Production 2 Building



Figure 2.53 - Conventional fire alarm control panel showing fault indication installed in the Security Room



Figure 2.54 - Faulty sounder on the first floor of the Composite & Textile Production Building



Figure 2.55 - Improperly placed manual call point due to excessive height in Electrical Workshop



Figure 2.56 - Non-fire rated cables used in fire alarm connection box for the fire alarm and detection system

9. Emergency Lighting

9.1 Provided but non-functional

The section refers to the entire facility.

During the inspection, it was observed that emergency lighting and illuminated exit signs were provided but not functional in a number of locations, particularly in the Management Building, Football Production 1 & 3 Buildings, Chemical Store, Press Cutting Hall, Generator Room 1, and Canteen.

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202603-064094

Timeline	Remediation Plan
Medium Term CAP-0075472	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency.
Longer Term CAP-0075473	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.57 - Non-functional emergency exit sign found in the Management Building



Figure 2.58 - Non-functional emergency light and exit light found in the Canteen



Figure 2.59 - Non-functional emergency light found on the first floor of Football Production 1



Figure 2.60 - Non-functional emergency light found in the Generator Room 1

9.2 Provided but insufficient

The section refers to the entire facility.

During the inspection, the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. Several areas lack sufficient emergency lighting and exit directional signs coverage. This issue was observed in the Football Production Buildings, Composite & Textile Production Building, Chemical Store Building, Electrical Workshop, Generator Room 1, and Welder shop. Additionally, the layouts for emergency lighting were missing. The absence of emergency lighting layouts throughout the facility raises concerns regarding the adequacy of illumination provided during power failures or emergencies.

ii. The backup power supply to the emergency luminaires was insufficient. The emergency light battery backup test failed on the stairs leading to the second floor of the Football Production 1 Building.

iii. External emergency lighting was not provided, particularly in the assembly area and external emergency stairs, preventing people from reaching a place of safety outside the facility during nighttime evacuations.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202603-064095

Timeline	Remediation Plan
Medium Term CAP-0075454	i. Conduct a comprehensive survey to identify areas lacking adequate emergency lighting. Prepare and submit the emergency light design and layouts to the Pakistan Accord for review. Install additional emergency luminaires and exit directional signs in these areas to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation.
Medium Term CAP-0075853	ii. Upgrade the backup power supply to ensure it can adequately support all emergency luminaires during a power failure. This includes installing additional battery packs or generators as necessary to comply with the Building Code of Pakistan Fire Safety Provisions 2016.

Medium Term CAP-0075455	iii. Install external emergency luminaires to enable safe evacuation during hours of darkness, all the way to a safe place outside the building(s).
Longer Term CAP-0075457	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.61 - Emergency light missing in the Welder Shop



Figure 2.62 - Low battery backup in emergency light at staircase to second floor, Football Production 1



Figure 2.63 - External emergency lights not provided in the emergency stairs of Chemical Store



Figure 2.64 - Emergency lighting and exit directional signs not installed in the Electrical Workshop



Figure 2.65 - Insufficient emergency lights coverage on the ground floor of Composite & Textile Production building



Figure 2.66 - Emergency lighting and exit directional sign not installed in the Generator Room 2

9.3 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected in a manner that would ensure automatic activation upon the loss of power to the general lighting in various spaces.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages. Further the emergency lights and exit signs should be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system. According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202603-064089

Timeline	Remediation Plan
Medium Term CAP-0075474	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately upon failure of general lighting power as required by the Building Code of Pakistan Fire Safety Provisions 2016. Emergency lighting shall be powered from a source independent of that supplying the normal lighting.
Longer Term CAP-0075475	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

9.4 General Lighting

The section refers to the entire facility.

During the inspection, the general lighting throughout the facility was assessed.

It was observed that the general lighting was controlled directly from the breakers within the Distribution Boards rather than through standard light switches. This issue was observed with the normal lights and boundary wall lights in Generator Room 1, on the ground floor of Football Production 1 Building, and on the first floor of the Composite & Textile Production Building.

Insufficient general lighting in several locations can increase the risk of accidents and impede safe evacuation during emergencies, while controlling general lighting via breakers instead of light switches poses operational inefficiencies and safety hazards. This setup can delay emergency responses and heighten the risk of electrical faults or fires due to frequent breaker use. Addressing these issues is crucial for enhancing everyday safety and ensuring effective emergency preparedness in the facility. F-202603-064098

Timeline	Remediation Plan
Medium Term CAP-0075452	Reconfigure the electrical control system to ensure that general lighting is operated via standard wall-mounted light switches instead of circuit breakers. This involves installing appropriate switches and rewiring the control circuits as per the National Electrical Code (NEC) and other relevant electrical safety standards. Ensure that all switches are easily accessible and clearly labelled to enhance operational efficiency and safety.



Figure 2.67 - General lighting controlled directly from the breakers on the first floor of the Composite & Textile Production



Figure 2.68 - General lighting controlled directly from the breakers in the Generator Room 1



Figure 2.69 - General lighting controlled directly from the breakers on the ground floor of Football Production 1

10. Generator, Transformer & Sub-Station Room

10.1 Ventilation and illumination

This section refers to the transformer room, sub-station rooms, and generator room.

An assessment of the existing ventilation system was conducted, alongside an evaluation of the lighting setup in various parts of the room, to determine their compliance with recommended standards.

i. The ventilation system was inadequate, with insufficient airflow leading to heat accumulation, particularly in the HT and LT rooms of the Electrical Workshop.

ii. One of the exhaust fans was found to be faulty in the Electrical Workshop.

iii. Several sections of the room were poorly lit, with light levels falling below the minimum required for safe operation and maintenance, particularly in the HT room of the Electrical Workshop.

BS7671 18th Edition + A2 (2022) specifies the need for ventilation to prevent overheating and ensure the safe operation of electrical installations. The standard also emphasizes that adequate lighting must be provided to ensure visibility for safe operation and maintenance. Proper lighting levels are crucial to avoid accidents and to ensure that any issues with the electrical installation can be promptly identified and addressed. F-202603-064096

Timeline	Remediation Plan
Medium Term CAP-0075468	i. Hire a qualified engineer to reassess the ventilation system to ensure it meets safety and performance standards. If the assessment concludes that the current system is inadequate redesign the ventilation layout to enhance the distribution of airflow, ensuring that critical areas around the HT panel and transformer receive adequate ventilation. This may involve repositioning vents or adding ductwork as necessary. Submit the ventilation assessment report and the proposed design to Pakistan Accord for review.
Medium Term CAP-0075469	ii. Repair any malfunctioning fans and replace those that are inoperative.

Medium Term CAP-0075470	iii. Add more lighting fixtures in poorly lit areas to ensure even distribution of light throughout the room. Focus on areas that require the highest visibility, such as around equipment and control panels.
Longer Term CAP-0075471	Establish regular maintenance and testing of the ventilation system and luminaires within the spaces to ensure adequate performance.



Figure 2.70 - Insufficient ventilation and illumination in HT room of Electrical Workshop



Figure 2.71 - Exhaust fan found faulty in HT room of Electrical Workshop

10.2 Working clearance

This section refers to the electrical switchrooms, generator room(s), and transformer room(s).

During our inspection, the working clearances surrounding key electrical equipment were assessed.

During the inspection, it was observed that the working clearance was inadequate in multiple areas. Equipment was placed too close to walls or other machines, making it difficult for personnel to access for routine maintenance or emergency interventions. Additionally, some pathways were obstructed, restricting movement and potentially delaying response times during an emergency. This issue was specifically observed with the LT panels installed in the Electrical Workshop and Generator Room 2.

Proper working clearance is crucial for ensuring life and fire safety. Adequate clearance allows personnel to safely access and maintain equipment, thereby reducing the risk of accidents or injuries caused by cramped conditions. Insufficient working clearance increases the risk of equipment failure due to limited maintenance access, which can lead to fires or other hazardous situations. F-202603-064100

Timeline	Remediation Plan
Medium Term CAP-0075403	Reorganize the equipment layout to ensure adequate clearance, remove obstructions in pathways, and implement clear floor markings and signage to designate working areas and pathways.
Longer Term CAP-0075404	Conduct regular workspace audits to maintain clearances, update maintenance protocols to include clearance checks, provide personnel training on maintaining clearances, and consider installing adjustable equipment to accommodate flexible positioning.



Figure 2.72 - Inadequate working clearance for LT panels installed in Generator Room 2



Figure 2.73 - Inadequate working clearance for LT panel installed in Electrical Workshop



Figure 2.74 - Inadequate working clearance for LT panels installed in Generator Room 2

10.3 Breathing system

This section refers to the transformer(s) serving the facility.

During the inspection of the transformer's breathing system, an assessment was conducted on its components, including the conservator tank, breather, and associated valves and piping. The condition of the silica gel or other desiccants within the breather was examined to ensure proper functionality.

It was observed that the silica gel in the breather was found to be saturated, indicating the need for replacement. This issue was observed with the transformer placed on the rooftop of Generator Room 2.

Maintaining an effective breathing system in a transformer is critical for fire and safety. It prevents mechanical failures by releasing excess pressure, thus reducing the risk of tank rupture. Furthermore, controlling moisture levels in the insulation oil minimizes the likelihood of electrical faults and potential fires, thereby enhancing overall safety measures.F-202603-064102

Timeline	Remediation Plan
Medium Term CAP-0075432	Replace the silica gel in the breather and repair any leaks around the breather housing to prevent moisture ingress into the transformer's insulation oil. Submit the transformer oil testing report to Pakistan Accord.
Longer Term CAP-0075433	Implement a regular maintenance schedule to periodically inspect and service the breathing system components, ensuring the silica gel is replaced as needed and that all seals and connections are intact to maintain system integrity.



Figure 2.75 - Silica gel in the breather was found to be saturated in the transformer placed on the rooftop of Generator Room 2

10.4 Room separation

During the inspection of the generator, transformer, and substation rooms, the physical separation between these critical areas was assessed. The barriers, walls, and partitions designed to isolate each component were examined to ensure they met safety standards and provided adequate separation.

It was observed that the HT and LT panels were located in close proximity without fire separation in the Electrical Workshop, posing a risk of fire propagation and potential impact on critical electrical systems.

Proper separation is essential to prevent incidents in one area from escalating and affecting adjacent sections, which can lead to widespread damage, injuries, or fatalities. Ensuring robust room separation enhances the overall safety and reliability of the facility. F-202603-064103

Timeline	Remediation Plan
Medium Term CAP-0075440	Install temporary fire-resistant barriers and seal any gaps and openings between the HT Panel and LT panel to immediately reduce the risk of fire and electrical fault propagation.
Longer Term CAP-0075441	Redesign and reconstruct the layout to include permanent fire-resistant walls and partitions that comply with relevant safety standards such as BS 7671 and NFPA 110, ensuring proper physical separation and enhanced safety for all critical areas. Establish a regular inspection and maintenance schedule to ensure the integrity of these barriers over time.



Figure 2.76 - HT and LT panels were located in close proximity without fire separation in Electrical Workshop

10.5 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

The following observations were noted with the battery system:

i. Noticeable spills of battery fluid were observed, which can lead to corrosion and pose a chemical hazard.

ii. Several battery terminals were exposed and not properly insulated, increasing the risk of electrical shock or short circuits.

iii. The placement of some batteries did not comply with safety regulations, potentially hindering safe access and proper ventilation.

These issues were observed with the batteries placed in the Management Building, Composite & Textile Production Building, R&D Lab, and Generator Rooms 1 & 2.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202603-064101

Timeline	Remediation Plan
Medium Term CAP-0075421	i. Clean any battery fluid spills using appropriate neutralizing agents and safety equipment to prevent corrosion and chemical hazards. Inspect and repair any affected battery cells or enclosures to prevent further leakage.
Medium Term CAP-0075422	ii. Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Medium Term CAP-0075423	iii. Reposition batteries that are improperly placed to ensure they comply with safety regulations, allowing for safe access and adequate ventilation. Mark and clear pathways around battery installations to ensure easy and safe access for maintenance and emergencies.
Longer Term CAP-0075424	i. Implement a regular maintenance schedule to monitor and address potential leaks, ensuring that any spills are promptly managed.

Longer Term CAP-0075425	ii. Establish a routine inspection and maintenance protocol to check and maintain proper insulation of all battery terminals regularly.
Longer Term CAP-0075426	iii. Conduct regular audits of battery placements to ensure ongoing compliance with safety standards and regulations.



Figure 2.77 - Exposed terminals observed on the UPS batteries located in the Management Building



Figure 2.78 - Exposed terminals observed on the generator batteries located in the Generator Room 2



Figure 2.79 - Fluid spillage and exposed terminals observed on the generator batteries located in the Generator Room 1



Figure 2.80 - Inappropriate placement for UPS batteries located in the Generator Room 2

11. Cabling & Wiring

11.1 Cable support and protection

This section refers to the entire facility.

During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

Proper supports were not provided for cables; they were laid directly on the ground and hung along the walls. Inadequate cable support results in sagging or stress on cables, which can lead to damage or failure.
This issue was specifically observed with the 11 kV cable between the electrical workshop and Generator Room 2, solar cables between Football Production 2 and Football Production 3 Buildings, and cables between Generator Room 2 and the Composite Production 2 ancillary structures. F-202603-064081

Timeline	Remediation Plan
Medium Term CAP-0075144	Secure cables with appropriate supports to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations).
Longer Term CAP-0075145	Implement a comprehensive cable management system with regular inspections to ensure cables remain properly supported and insulated over time.



Figure 2.81 - Cables laid without proper support between Generator Room 2 and Composite Production 2



Figure 2.82 - 11 kV cable without proper support between Electrical Workshop and Generator Room 2



Figure 2.83 - Solar cables improperly supported between Football Production 2 and 3 Buildings

11.2 Cable trench and channel cleanliness

This is related to the whole facility.

The following observations were noted during the inspection of the cable trunking systems:

i. The cleanliness of the cable trunking system had not been maintained. The trunking systems were found to be filled with a significant accumulation of fluff and dust, posing a potential fire hazard. This issue was observed in the cable trenches of Generator Rooms 1 and 2 and in the cable trays on the first floor of Football Production 1 and Composite & Textile Production Buildings.

ii. Power cables and compressed air pipes were routed together in electrical cable trays in the stitching hall, posing a potential safety risk due to moisture from air leakage or condensation, and possible mechanical damage to the cables. This issue was observed in the cable trays on the first floor of the Composite & Textile Production Building.

Accumulated dust and fluff can ignite when exposed to sparks or high temperatures, potentially leading to a fire hazard. Maintaining a clean cable trunking system is essential to minimize fire risks associated with accumulated debris.F-202603-064078

Timeline	Remediation Plan
Medium Term CAP-0075295	i. Clean all cable trunking systems to remove accumulated dust and fluff, ensuring that any potential fire hazards are mitigated.
Medium Term CAP-0075296	ii. Ensure that power cables and compressed air pipes are routed separately in accordance with relevant safety codes and standards to prevent potential hazards.
Longer Term CAP-0075297	Implement a regular maintenance schedule that includes routine inspections and cleaning of the cable trunking systems to prevent future accumulation of dust and fluff, thereby maintaining a safe environment and complying with safety standards.



Figure 2.84 - Cable trench filled with dust and debris found in the Generator Room 2



Figure 2.86 - Lint and dust inside the cable tray on the first floor of Football Production 1



Figure 2.85 - Power cables and compressed air pipes routed together inside the cable tray on the first floor of Composite & Textile Production Building



Figure 2.87 - Cable trench filled with dust and debris found in the Generator Room 1

11.3 Splicing of cables

This relates to the whole facility.

During the inspection, the cable installation was visually surveyed throughout the facility.

The following observations were noted:

i. Cables terminations within panels and switchgears were made without cable lugs, resulting in broken strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard. This issue was observed in the LT panel and electrical enclosures of Generator Room 1 and in multiple distribution boards, particularly in the padel racket area and stitching section of the Composite & Textile Production Building and the Composite Production 2 Building.

ii. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks. This issue was specifically observed in the textile main distribution board on the ground floor and the stitching distribution board on the first floor of the Composite & Textile Production Building, as well as the main distribution board of the Composite Production 2 Building.

iii. Non-compliant PVC connectors used in distribution boards, which could pose a hazard of a potential short circuit due to their poor insulation. This issue was specifically observed in the distribution boards and electrical enclosures of the water turbine in Generator Room 2, the drilling machine in the Welder Shop, and the CNC drilling machine in the Composite Production 2 Building.

For any future electrical work, the termination of stranded cables must be done with cable lugs, and PVC tape joints shall not be utilized. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety. F-202603-064077

Timeline	Remediation Plan
Medium Term CAP-0075317	i. Install cable lugs on all improperly terminated cables within panels and switchgears to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022).

Medium Term CAP-0075318	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring.
Medium Term CAP-0075319	iii. Replace the PVC connector with a standard quality terminal block in accordance with BS7671 18th Edition + A2 (2022).
Longer Term CAP-0075320	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.
Longer Term CAP-0075321	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.
Longer Term CAP-0075322	iii. Establish a plan to replace all low-quality PVC connectors with compliant electrical terminal blocks and integrate this into regular maintenance to ensure ongoing safety and compliance.



Figure 2.88 - Improper cable joints and non-compliant PVC connector in drilling machine, Welder Shop



Figure 2.89 - PVC tape joints and connector found in the main distribution board of Composite Production 2



Figure 2.90 - Spliced cables covered with PVC tape found in the textile main distribution board of Composite & Textile Production



Figure 2.91 - Cable terminations without lugs PVC tape joints found in stitching distribution board on the first floor of Composite & Textile Production Building

11.4 Colour coding

This relates to the whole facility.

During the inspection of several distribution boards, deviations from standard cable colour coding practices were identified. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires.
This issue was observed specifically on the wooden hockey production distribution board in the Composite Production 1 Building, the football printing distribution on the second floor in the Football Production 1 Building, and the football molding distribution board on the first floor in the Football Production 3 Building.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202603-064082

Timeline	Remediation Plan
Medium Term CAP-0075293	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term CAP-0075294	Develop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.

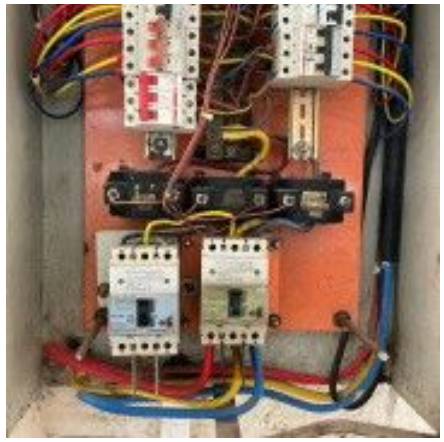


Figure 2.92 - Incorrect color coding found in the football molding distribution board on the first floor of Football Production 1 Building

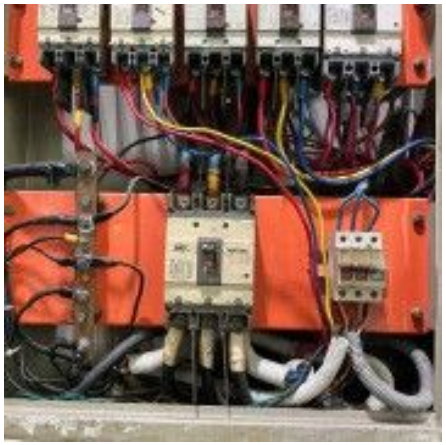


Figure 2.93 - Incorrect color coding found in the wooden hockey production distribution board of Composite Production 1 Building



Figure 2.94 - Incorrect color coding found in the football printing distribution board on the second floor of Football Production 1 Building

11.5 Fixtures

This relates to the whole facility.

During the inspection, the electrical fixtures throughout the facility were visually surveyed.

The following observations were noted:

i. Power receptacles installed in inappropriate locations: Receptacles placed in unsuitable locations, such as non-IP-rated sockets installed outdoors or exposed to the elements, are at higher risk of moisture ingress and corrosion. This can lead to electrical short circuits or fires, posing a severe safety hazard. This issue was specifically observed outside the Compressor Room, Electrical Workshop, Gas Steamer Area, and Generator Rooms 1& 2.

ii. Power receptacles mounted on combustible wooden ply found in the Welder Shop. Installing receptacles on combustible materials like wooden ply increases the risk of fire if a short circuit or overheating occurs. Combustible materials can easily ignite, leading to potential fire hazards and endangering life safety.

iii. Broken, burnt, and exposed power receptacles found in the Security Office and outside the Electrical Workshop. Damaged receptacles with exposed parts can lead to loose connections and increased electrical resistance. This condition can cause overheating, potentially igniting surrounding materials and increasing the risk of electrical fires, posing a serious life safety threat.

iv. Multiple stitching machine plugs were connected to a single socket, posing a risk of overloading, overheating, and potential fire. Observed in the textile stitching on the first floor of the Composite & Textile Production Building.F-202603-064076

Timeline	Remediation Plan
Medium Term CAP-0075356	i. Relocate power receptacles from inappropriate locations to more suitable, protected areas to prevent exposure to environmental elements. For external receptacles utilize suitably rated IP fixtures.
Medium Term CAP-0075357	ii. Replace combustible wooden ply backings with non-combustible materials to reduce fire risk.

Medium Term CAP-0075358	iii. Replace or repair broken and exposed power receptacles immediately to restore safe and secure connections.
Medium Term CAP-0075359	iv. Disconnect excess plugs and ensure each machine is connected to a dedicated socket of appropriate rating.
Longer Term CAP-0075360	i. Develop installation guidelines and training for personnel to ensure future power receptacles are installed in compliant and safe locations.
Longer Term CAP-0075361	ii. Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance.
Longer Term CAP-0075398	iii. Implement a preventive maintenance program that includes periodic checks and timely replacement of damaged electrical components to maintain safety standards.
Longer Term CAP-0075399	iv. Establish and implement a facility-wide standard to ensure that all equipment is connected to appropriately rated, dedicated power outlets without overloading sockets.





Figure 2.95 - Burnt and non-IP rated power receptacle installed outside Electrical Workshop



Figure 2.96 - Non-IP rated power receptacle installed outside the compressor area



Figure 2.97 - Burnt out power receptable found in the Security Room



Figure 2.98 - Power receptacle installed with wooden backing in the Welder Shop

Figure 2.99 - Multiple stitching machine plugs connected to a single socket on the first floor of the Composite & Textile Production Building

12. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.